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RESEARCH ARTICLE

Halogenated Benzothiadiazole Solid Additives Enable 20.12% Efficiency Layer-by-Layer Organic Solar Cells

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ABSTRACT

Small molecule solid (SMS) additives have been widely used to optimize active layer morphology for improving power conversion efficiencies (PCEs) of organic solar cells (OSCs). Herein, three halogenated benzothiadiazole derivatives (named 2*X*BT, including difluorinated 2*F*BT, dichlorinated 2*Cl*BT, and dibrominated 2*Br*BT) were developed as SMS additives to boost the photovoltaic performance of the OSCs based on layer-by-layer (LBL) processed D18/L8-BO active layers. With the halogen atomic mass increases, three 2*X*BT additives exhibit progressively strong intermolecular interactions with active layer materials (D18 and L8-BO). As a result, the 2*Br*BT-treated LBL active layer achieves an optimized blend morphology with ideal molecular crystallinity and vertical phase separation promoting exciton dissociation and charge transfer, while suppressing charge recombination in OSCs, yielding an impressive PCE of 19.15%. Moreover, by introducing a near-infrared absorbing guest acceptor of eC9-4ClO into the D18/L8-BO host to broaden absorption, the OSCs based on 2*Br*BT-treated ternary D18/L8-BO:eC9-4ClO obtained a further boosted PCE of 20.12% with an increased photocurrent. Notably, this PCE of 20.12% represents the highest value among the reported LBL processed OSCs. This work highlights that the halogenation in benzothiadiazole derived SMS additives is a promising strategy to optimize active layer morphology for obtaining efficient OSCs.

Muhammad Hamza Maqsood, Na Chen, Kai Xiang, and Hairui Bai equally contributed to this work.

1 | Introduction

Organic solar cells (OSCs) have attracted huge attention owing to their excellent mechanical flexibility, lightweight, and large-area roll-to-roll manufacturing [1–8]. Thanks to the constant advancements in developing new high-performance photovoltaic materials, particularly with regard to narrow bandgap non-fullerene small molecule acceptors (SMAs) [9–13] and wide bandgap polymer donors [14–18], the state-of-the-art OSCs have attained power conversion efficiencies (PCEs) exceeding 20% [19–25], surpassing the threshold for industrial applications. Apart from the continuous innovation of photovoltaic materials, the optimization of active layer morphology is also crucial for further improving the photovoltaic performance of OSCs [26–28]. The active layers with bulk heterojunction (BHJ) morphology characteristics, as a key of the photon-to-electron conversion in OSCs, should have ideal vertical phase separations between donor and acceptor components and a superior continuous double-fibril network structure, to achieve efficient exciton dissociation and charge transport properties [29, 30]. Nevertheless, due to the structural skeleton homology between many donor and acceptor materials, their Flory–Huggins interaction parameter (χ) is typically small, leading to high compatibility, which may confine the formation of ideal vertical phase separations [31]. Moreover, the distinct self-aggregation and crystallization kinetics of donor and acceptor materials lead to the complex molecular stacking, making it challenging to finely control the morphology of active-layers for forming continuous double-fibril interpenetrating networks [32]. Therefore, the precise optimization of active layer morphology remains one of the challenges in the fabrication of efficient OSCs.

Compared with BHJ active layers fabricated by using a traditional one-step deposition method, the quasi-planar heterojunction (Q-PHJ) active layers prepared by a layer-by-layer (LBL) technology tend to form improved vertical phase distribution, mainly due to that the donor and acceptor components are deposited separately [33–36]. However, the Q-PHJ active layers inherently provide limited donor/acceptor interfaces, while they can be optimized by using distinct solvents and additives to enhance interlayer penetration [37]. Among the above-mentioned treatments, adding appropriate small molecule solid (SMS) additives has become one of the most successful approaches for morphology optimization, as they have the advantages of cost-effectiveness, simple use, and efficient manipulation of active layer morphology [38–44]. High-performance SMS additives usually have strong non-covalent interactions with both/separate donor and acceptor components, as well as the special capacity to combine thermal annealing induced molecular recrystallization and volatilization-driven intermolecular re-stacking [45]. This can provide some critical benefits, such as inducing mutual permeation of photovoltaic materials to increase the donor/acceptor interface, improving intermolecular stacking, and optimizing phase separation. Therefore, these relevant OSCs can achieve superior double-fibril interpenetrating networks to boost exciton dissociation and charge transport for offering high short-circuit current density (J_{SC}) and fill factor (FF), as well as regulate the energy level difference between the excited state of acceptor component and the charge transfer state in mixed phases, for minimizing voltage loss [46, 47].

In term of their molecular structures, the normally used SMS additives can be broadly catagorized into the following: i) conjugated/non-conjugated SMS additives [48–55], ii) conformational lock typed SMS additives [56, 57], iii) planar/3D structured SMS additives [58, 59], and iv) optical functional SMS additives [60–62], et al. (Figure S1). Despite considerable progress has been made in improving PCEs using these additives, their applications in the LBL processed OSCs have been much less explored. On the other hand, high-performance SMS additives specifically designed for LBL-processed OSCs are scarce. Benzothiadiazole (BT) unit, with the properties of strong electron-deficient capacity and easily molecular modification, has been widely used in designing and synthesizing high-performance wide bandgap polymer donors (such as D18 [15]), near-infrared absorbing Y-series SMAs (Y-SMAs, e.g Y6 [15] and L8-BO [10]), and other organic semiconductor materials. In addition, the bromine (*Br*) atom has been widely incorporated to construct high-performance photovoltaic materials, mainly due to that their strong electronegativity and empty valence orbitals can form strong intramolecular non-covalent bonds and intermolecular interactions [63, 64]. Therefore, introducing *Br* atom into BT unit tends to effectively adjust its charge distributions and thermodynamic sublimation, which opens an opportunity to manipulate Q-PHJ active layer morphology. However, although the BT unit has been used as a SMS additive to boost photovoltaic performance of OSCs, brominated BT additives that can better regulate Q-PHJ active layer morphology have not yet been reported [65].

Herein, three dihalogenated SMS additives named 2*X*BT (including difluorinated 2*F*BT, dichlorinated 2*Cl*BT, and dibrominated 2*Br*BT) were developed by attaching different halogen atoms onto their parental halogen-free BT unit to optimize the active layer morphology of LBL processed D18/L8-BO. As the halogen atomic mass increases, three 2*X*BT additives show gradually increased intermolecular interactions with active layer materials of both D18 and L8-BO components. Therefore, the 2*Br*BT-treated D18/L8-BO active layer demonstrates better active layer morphology with superior intermolecular packing and optimal vertical phase separation compared with additive-free, 2*F*BT, and 2*Cl*BT-treated counterparts, which is beneficial for obtaining the improved exciton dissociation and charge transport properties. As a result, the 2*Br*BT-treated OSCs yielded an excellent PCE of 19.15%, surpassing these of ones treated with additive-free (18.45%), 2*F*BT (18.78%), and 2*Cl*BT (18.63%), respectively. Moreover, using a near-infrared absorbing guest acceptor (eC9-4ClO) into D18/L8-BO host to broaden absorption, the 2*Br*BT treated ternary OSCs obtained a further boosted PCE of 20.12% with a significantly increased J_{SC} from 26.86 to 27.70 mA cm^{-2}. Notably, this 20.12% efficiency is the highest value among the reported LBL processed OSCs. This work highlights that halogenation of BT derived SMS additives is a promising strategy to optimize active layer morphology for achieving efficient OSCs.

2 | Results and Discussion

Molecular structures of active layer materials (D18 and L8-BO) and 2*X*BT additives (2*F*BT, 2*Cl*BT, and 2*Br*BT) are shown in

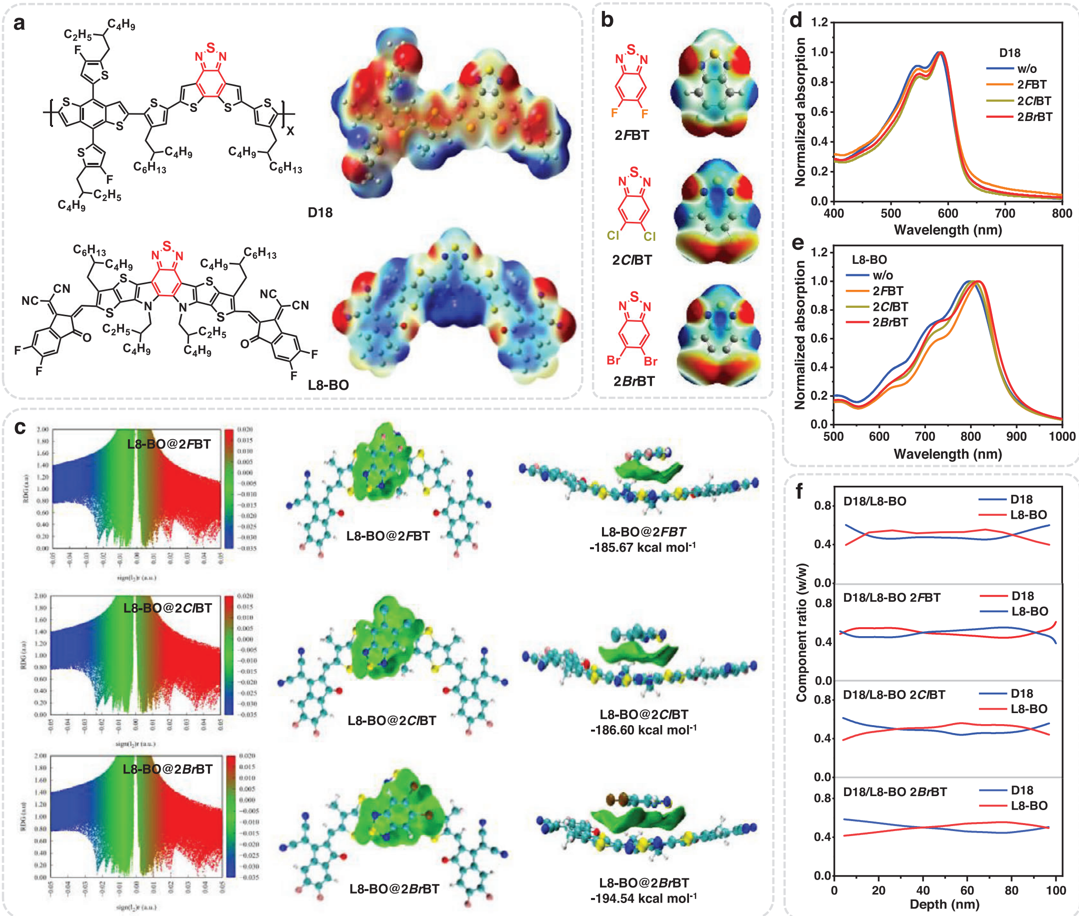

FIGURE 1 | Molecular structures and related ESP patterns of (a) active layer materials and (b) 2*X*BT additives, respectively. (c) The functions between RDG and Sign(λ_2)ρ, and the corresponding RDG *iso*-surface maps of L8-BO@2*X*BT pairs, respectively. Normalized absorption spectra of (d) D18 and (e) L8-BO neat films treated without/with 2*X*BT additives, respectively. (f) Film-depth-dependent component distributions of D18 and L8-BO components in D18/L8-BO active layers treated without/with 2*X*BT additives.

Figure 1a,b. With the halogen atomic mass increases, 2*X*BT additives offer increased melting points from 66°C to 112°C and 131°C, implying that 2*Br*BT has a longer intermolecular interaction time during film-forming. As shown in Figure S2, the volatility of the 2*X*BT additives was investigated, wherein all of them can be volatilized rapidly and completely under 100°C for 8 min.

As shown in Figure 1d, compared with the additive-free D18 film, the 2*X*BT-treated ones depict almost the same absorption spectra but slightly enhanced absorption intensities belonging to *J*-aggregation, indicating enhanced intermolecular ordered stacking. In contrast, after introducing 2*X*BT additives, the L8-BO neat films exhibit obviously red-shifted absorption peaks and onsets, suggesting enhanced intermolecular π-π packing. A similar phenomenon has been also found in the 2*Br*BT-treated D18/L8-BO blends. As displayed in Figure S3, the 2*Br*BT treated D18/L8-BO blends achieved higher absorption coefficients of both the donor and acceptor components ($0.81 \times 10^5/0.89 \times 10^5$ cm^{-1}) compared with those of the additive-free ($0.66 \times 10^5/0.82 \times 10^5$ cm^{-1}), 2*F*BT ($0.78 \times 10^5/0.87 \times 10^5$ cm^{-1}), and 2*Cl*BT treated ones ($0.71 \times 10^5/0.85 \times 10^5$ cm^{-1}), which is beneficial for photons harvesting. The highest occupied molecular orbital (HOMO) and lowest unoccupied molecular orbital (LUMO) energy levels of D18, L8-BO, and eC9-4ClO were summarized in Figure S4, which were obtained from literature [10, 15].

Electrostatic potentials (ESP) of PM6, L8-BO, and 2*X*BT were calculated by using density functional theory (DFT). As depicted in Figure S5, for the BT unit, its positive charges dominate and concentrate at the benzene ring, and negative charges are mainly dispersed on the thiadiazole moiety. When attaching halogen atoms into the benzene ring, the 2*X*BT additives depict negative charges dominated backbones and has separated positive and negative charge distributions (Figure 1b). Therefore, unlike halogen-free BT unit, the 2*X*BT additives display relatively more balanced charge distributions in the thiadiazole moiety and benzene ring. According to the principles of antipolar attraction theory [66], halogenated 2*X*BT can offer strong non-covalent interactions with both donor (D18 with a negative charge-dominated distribution) and acceptor (L8-BO with an almost positive charge distribution across the entire backbone) components. As shown in Figure S6, the 2*F*BT, 2*Cl*BT, and 2*Br*BT additives offer the dipole moments of 1.10, 1.05, and 0.82 D, respectively. Therefore, 2*Br*BT with a lower dipole moment and more balanced positive and negative charge distributions tends to obtain improved non-covalent interactions with both donor and acceptor components. Besides, compared with *F* and *Cl* atoms, the *Br* atom has a larger atomic radius, which can improve spin-orbit coupling in π-conjugated system and strengthen their interaction [67].

A reduced density gradient (RDG) approach was used to examine the non-covalent interactions of the D18@2*X*BT and L8-BO@2*X*BT pairs. As depicted in Figure S7 and Figure 1c of the functions between RDG and Sign(λ_2)ρ plot, the blue zone indicates a strong attractive interaction, the red zone implies a strong steric effect, and the green zone suggests a weak van der Waals interaction, respectively [68]. All the D18@2*X*BT pairs exhibited similar strong attractive interactions and steric effects, in the RDG and visual representation maps. The similar phenomena were also observed in the L8-BO@2*X*BT pairs. Differently, the green zone representing weak van der Waals interactions slightly increases in D18 (or L8-BO) paired sequentially with 2*F*BT, 2*Cl*BT, and 2*Br*BT, suggesting a steady improvement in non-covalent interactions. Moreover, the binding energy values of D18@2*X*BT and L8-BO@2*X*BT pairs were calculated. Among them, the D18@2*Br*BT and L8-BO@2*Br*BT pairs offered much lower binding energy values, indicating stronger intermolecular interactions, which favor tighter and more ordered intermolecular stacking of active layer materials.

To further investigate the impact of 2*X*BT additives on the vertical phase separation of active layers, the film-depth-dependent light absorption spectroscopy (FLAS) measurement was employed (Figure S8a) [69, 70]. By fitting the FLAS curves with the individual absorption profiles of each component, the depth-dependent composition distributions of D18 and L8-BO at different film-depths were extracted (Figure 1f). Owing to the superior non-covalent interactions in both D18@2*Br*BT and L8-BO@2*Br*BT pairs, the 2*Br*BT treated active layer offered more balanced donor/acceptor distributions of D18 and L8-BO in the 20–90 nm depth range compared with these of active layers treated with additive-free, 2*F*BT, and 2*Cl*BT-treated films, indicating more favorable vertical phase separation. As shown in Figures S8b and S9, the exciton distribution within the active layers was also simulated using a transfer-matrix optical method. According to the exciton generation contours, the simulated exciton generation rate (G) at different film-depths were estimated. Due to the more balanced distributions of D18 and L8-BO components, the 2*Br*BT-treated films gained a higher G value of 19.39×10^{28} than those of films processed with additive-free (17.51×10^{28}), 2*F*BT (18.52×10^{28}), and 2*Cl*BT (17.68×10^{28}).

To study the effect of 2*X*BT additives on the photovoltaic performance of OSCs, the devices with a conventional structure of ITO/2PACz/active layer/PDINN/Ag were fabricated, where the active layers were prepared using an LBL method. The corresponding visual photographs of devices were presented in Figure S10. The current density-voltage (J-V) curves of the optimized OSCs based on D18/L8-BO processed without/with 2*X*BT additives under a simulated AM 1.5G solar irradiation (100 mW cm^{-2}) are presented in Figure 2a, and the related photovoltaic parameters are listed in Table 1. The additive-free OSCs produced an open-circuit voltage (V_{OC}) of 0.900 V, J_{SC} of 25.97 mA cm^{-2}, and FF of 78.96%, leading to a moderate PCE of 18.45%, which is comparable to the previously reported results [10]. When introducing the 2*X*BT additives to optimize active layer morphology, all the OSCs offered increased J_{SC} (26.47–26.86 mA cm^{-2}) and FF (79.12%–80.66%) values, resulting in boosted PCEs (18.63%–19.15%). Among the 2*X*BT-processed devices, the 2*Br*BT-treated OSC gained a lower V_{OC} but both higher J_{SC} and FF values, achieving a champion PCE, which is mainly due to its superior active layer morphology arising from the enhanced intermolecular interaction between 2*Br*BT and active layer materials. Since 2*Br*BT with a higher molecular weight has a longer intermolecular interaction time during film-forming, the D18/L8-BO based devices were further fabricated by increasing additive contents of 2*F*BT and 2*Cl*BT to prolong the film-forming time of active layers. As shown in Figure S11 and Table S1, with the increases of 2*F*BT and 2*Cl*BT contents (from 2.5 to 5.0, and then to 10.0 mg mL^{-1}), the corresponding OSCs achieved decreased PCEs. Moreover, a near-infrared absorbing guest acceptor eC9-4ClO was used to broaden the absorption of D18/L8-BO host. Ternary OSCs based on 2*Br*BT treated D18/L8-BO:eC9-4ClO achieved a further increased J_{SC} of 27.70 mA cm^{-2}, leading to an impressive PCE of 20.12% (Figure 2b). Notably, the PCE of 20.12% is the record-high value among the reported LBL processed OSCs (Table S2).

To verify the reliability of photovoltaic performance, the external quantum efficiency (EQE) curves of the OSCs were measured. As shown in Figure 2c, the OSCs based on 2*Br*BT treated D18/L8-BO and D18/L8-BO:eC9-4ClO yielded the integrated J_{SC} values of 26.04 and 26.64 mA cm^{-2} from their EQE curves, respectively, which well match with their J--V measurements.

The photocurrent density versus effective voltage (J_{ph}-V_{eff}) curves were plotted to explore the exciton and charge kinetics of OSCs. The exciton dissociation (η_d) and charge collection (η_c) probabilities of devices were calculated by the ratio of J_{ph} and saturation current density (J_{sat}) under short-circuit condition and maximal power output condition, respectively. As shown in Figure 2d,h, the η_d and η_c values were determined to be 96.1%/78.1%, 97.1%/80.8%, 96.7%/78.9%, and 97.6%/81.3% for the devices based on D18/L8-BO processed with additive-free, 2*F*BT, 2*Cl*BT, and 2*Br*BT, respectively. The superior η_d and η_c values in the 2*Br*BT treated device indicate improved exciton dissociation and charge collection properties, contributing to the increased J_{SC} and FF values.

To further understand the effect of the 2*X*BT additives on the photophysical process of OSCs, time-resolved photoluminescence (TRPL) measurement was performed. As exhibited in Figure 2e, the TRPL spectra of the additive-free, 2*F*BT, 2*Cl*BT, and 2*Br*BT-treated D18/L8-BO blends offered the photoluminescence lifetimes of 0.68, 0.59, 0.62, and 0.57 ns, respectively. The shorter TRPL decay lifetime of the 2*Br*BT treated blend suggests more efficient exciton dissociation in devices.

The charge carrier lifetime and charge extraction time of OSCs were further studied by measuring their transient photovoltaic (TPV) and transient photocurrent (TPC). In the fitted TPV decay curves (Figure 2f), the additive-free, 2*F*BT, 2*Cl*BT, and 2*Br*BT treated D18/L8-BO devices achieved gradually increased carrier lifetimes of 15.2, 17.6, 16.8, and 18.5 μs, respectively. The longer carrier lifetime for the 2*Br*BT treated device is beneficial for suppressing charge recombination. By fitting the TPC decay curves (Figure 2g), the charge extraction times of the

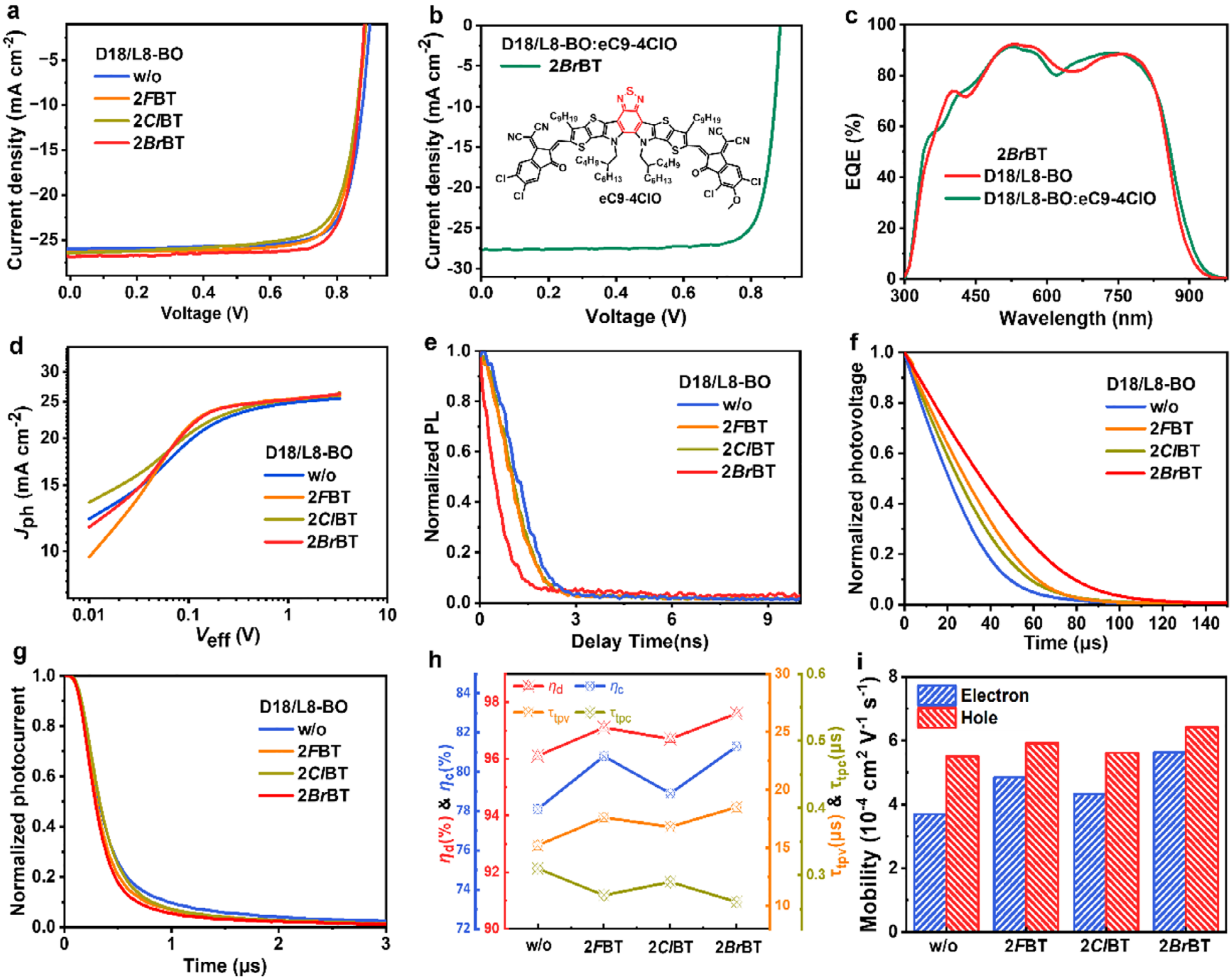

FIGURE 2 | *J–V* curves of the OSCs based on (a) D18/L8-BO processed without/with 2*X*BT additives and (b) D18/L8-BO:eC9-4ClO processed with 2*Br*BT additive. (c) EQE plots of the OSCs based on 2*Br*BT-treated D18/L8-BO and D18/L8-BO:eC9-4ClO. (d) J_{ph}-V_{eff} curves, (e) TRPL decay kinetics, (f) TPV decay kinetics, and (g) TPC decay kinetics of the OSCs based on D18/L8-BO processed without/with 2*X*BT additives, respectively. (h) Diagrams of carrier-related parameters. (i) Summarized μ_e and μ_h values of the devices based on D18/L8-BO processed without/with 2*X*BT additives.

TABLE 1 | Photovoltaic data of the OSCs in this work.

Active layer	Additive	V_{OC} [V]	J_{SC} [mA cm^{-2}]	FF [%]	PCE (ave. PCE)[a] [%]
D18/L8-BO	w/o	0.900	25.97	78.96	18.45 (18.24 ± 0.20)
D18/L8-BO	2*F*BT	0.888	26.47	79.90	18.78 (18.56 ± 0.18)
D18/L8-BO	2*Cl*BT	0.889	26.49	79.12	18.63 (18.41 ± 0.21)
D18/L8-BO	2*Br*BT	0.884	26.86	80.66	19.15 (18.96 ± 0.19)
D18/L8-BO:eC9-4ClO		0.885	27.70	82.07	20.12 (19.92 ± 0.17)

[a]The average PCE values and the corresponding standard deviations were achieved from 10 independent devices.

additive-free, 2*F*BT, 2*Cl*BT, and 2*Br*BT treated D18/L8-BO devices can be estimated as 0.31, 0.27, 0.29, and 0.26 μs, respectively. The faster charge extraction of the 2*Br*BT treated-device is beneficial for obtaining both higher J_{SC} and FF.

The charge transport properties of the 2*X*BT treated D18/L8-BO devices were further evaluated using the space-charge-limited current (SCLC) method. As detailed in Figure 2i, Figure S12, and Table S3, the devices based on D18/L8-BO processed with additive-free, 2*Cl*BT, 2*F*BT, and 2*Br*BT exhibited the hole/electron mobilities (μ_h/μ_e) of 5.51 × 10^{-4}/3.70 × 10^{-4}, 5.62 × 10^{-4}/4.32 × 10^{-4}, 5.92 × 10^{-4}/4.85 × 10^{-4}, and 6.42 × 10^{-4}/5.63 × 10^{-4} cm^2 V^{-1} s^{-1}, with the corresponding μ_h/μ_e ratios of 1.48, 1.30, 1.22, and 1.14, respectively. Among these devices, the 2*Br*BT-treated one offered

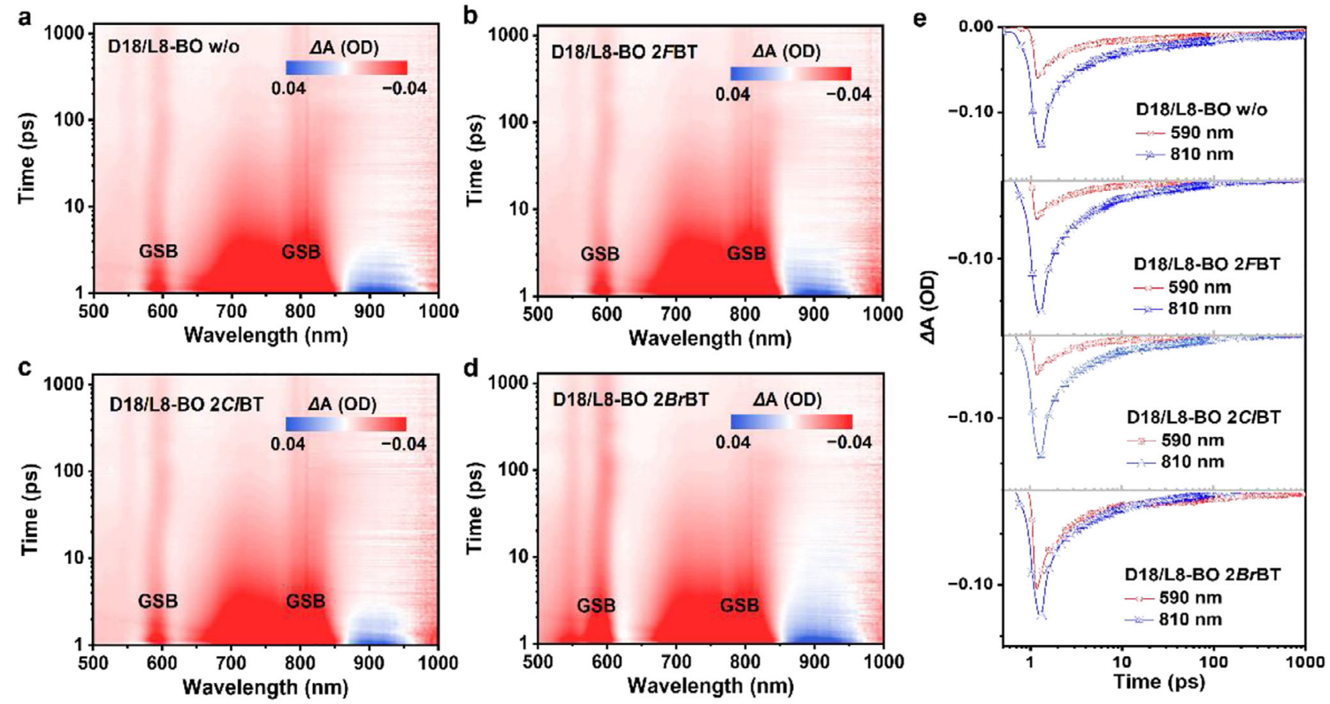

FIGURE 3 | (a–d) 2D color fs-TAS profiles of the D18/L8-BO blends processed without/with different 2*X*BT additives, and (e) the corresponding kinetics curves.

higher and more balanced charge transport properties, which also partly explain its higher J_{SC} and FF values.

Femtosecond transient absorption spectroscopy (fs-TAS) was measured to study the effect of the 2*X*BT additives on the photogenerated exciton dynamics. As shown in Figure 3a–d, a strong excited state absorption (ESA) peak at ~810 nm was observed immediately following excitation in all the blends. Then, the decay of the feature signal with a new ground state bleaching (GSB) peak appears at 590 nm, indicating the hole transfer from L8-BO to D18. As depicted in Figure 3e, the electron transfer kinetics and recombination process within active layers were numerically described, via fitting the GSB signal at 590 nm, using a biexponential function with two lifetimes (τ_1 and τ_2). The τ_1 is assigned to the ultrafast dissociation time at the D/A interfaces of the excitons formed in the acceptor component, while the τ_2 is related to the time for excitons moving to the D/A interfaces. The 2*Br*BT treated D18/L8-BO gained shorter τ_1 and τ_2 values (3.41 and 26.7 ps) compared with the additive-free (3.87 and 29.8 ps), 2*F*BT (3.55 and 27.7 ps), and 2*Cl*BT (3.67 and 28.6 ps) treated blends, suggesting faster exciton dissociation and more efficient exciton diffusion, which is in accordance with its superior J_{SC} and FF of the corresponding OSCs.

Atomic force microscopy (AFM) measurement was performed to gain a deeper understanding of how the 2*X*BT additives affect the surface morphology of D18/L8-BO blends. As shown in their AFM height images (Figure 4a), the 2*Br*BT treated D18/L8-BO film shows a more textured microstructure with a root-mean-square (RMS) roughness value of 3.75 nm, which is less than that of additive-free film (4.90 nm) but slightly high than that of 2*F*BT (2.81 nm) and 2*Cl*BT (2.60 nm) treaded films. Among their phase images (Figure 4b), the 2*Br*BT treated D18/L8-BO film also exhibits relatively regular and more obvious fibrillar interpenetrating network morphology with better homogeneity, which effectively encourages its charge generation, transport, and extraction properties.

Grazing-incidence wide-angle X-ray scattering (GIWAXS) measurement was utilized to further investigate molecular crystallinity and packing properties of 2*X*BT treated L8-BO neat films and D18/L8-BO blend films. As depicted in Figure 4c and Figure S13, the additive-free L8-BO neat film exhibits distinct (010) diffraction peaks in both the out-of-plane (OOP) and in-plane (IP) directions, suggesting the mixed "edge-on" and "face-on" orientations. Differently, the 2*X*BT treated L8-BO films offer both stronger (010) diffraction peaks in the OOP direction and (100) diffraction peaks in the IP direction, implying a predominantly "face-on" orientation. In the OOP direction, the (010) diffraction peaks corresponding to the intermolecular π–π stacking of the 2*X*BT treated L8-BO films are located at 1.78–1.82 Å^{-1}, with their π-π stacking distances of 0.34–0.35 nm and crystal coherence lengths (CCLs) of 1.62–1.87 nm. The GIWAXS results of D18/L8-BO blends processed without/with 2*X*BT additives are depicted in Figure 4d–e. Compared with the additive-free D18/L8-BO blend, the 2*X*BT treated D18/L8-BO blends display more prominent (100) lamellar stacking diffraction peaks at ~0.40 Å^{-1} in the IP direction with higher CCLs of 4.51–11.64 nm. In the OOP direction, the (010) diffraction peaks of the 2*X*BT treated D18/L8-BO films are located at 1.74–1.80 Å^{-1}, with the corresponding π-π stacking distances of 0.35–0.36 nm and CCLs of 1.84–1.92 nm. The above-mentioned results indicate that introducing 2*X*BT additives into D18/L8-BO active layer improves molecular crystallinity and packing properties, thereupon forming multilevel electron transport pathways, which also partly explain the simultaneously improved J_{SC} and FF values in OSCs.

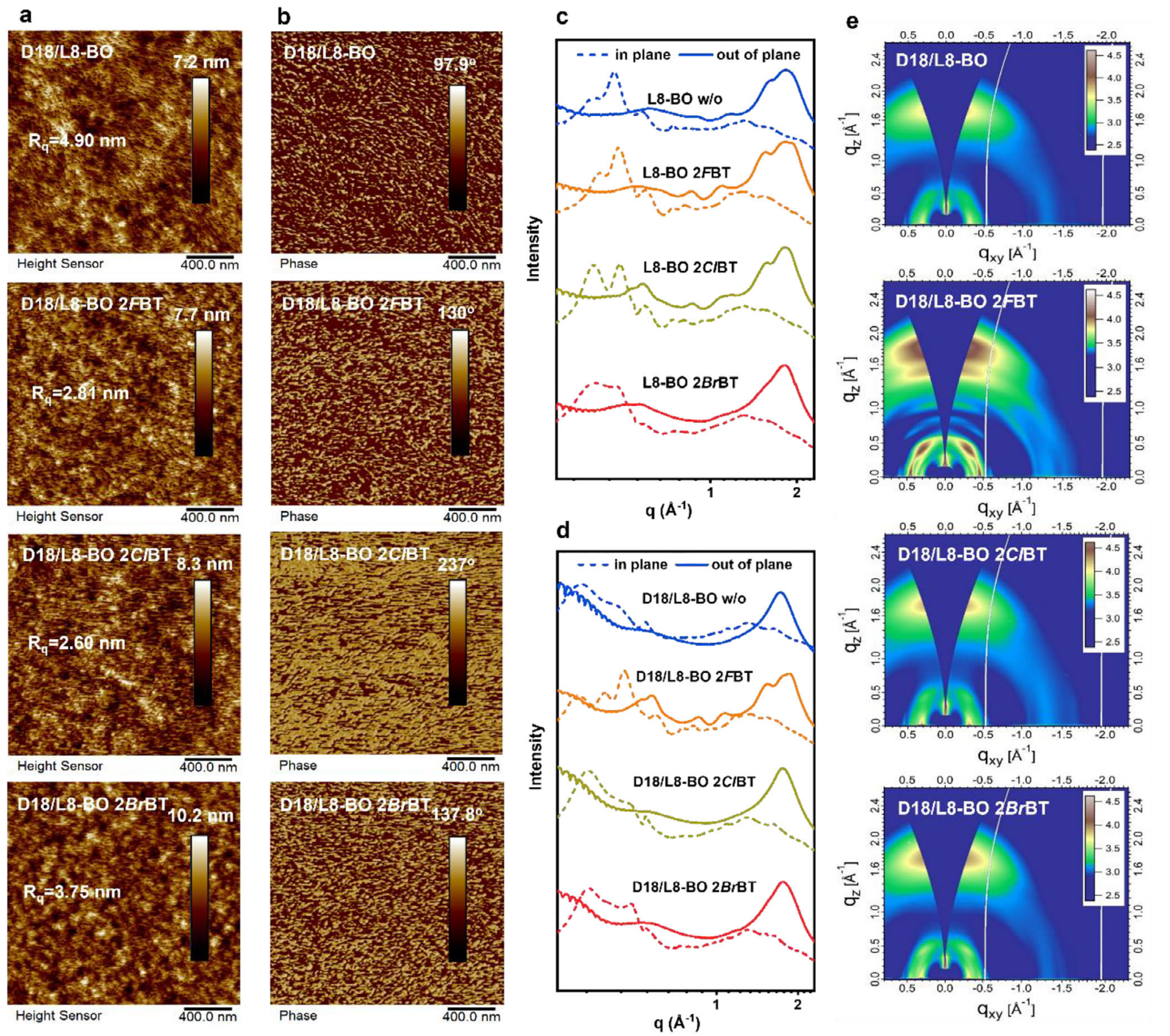

FIGURE 4 | (a) AFM height images and (b) AFM phase images of D18/L8-BO active layers processed without/with different 2*X*BT additives, respectively. Line-cutting curves obtained from 2D GIWAXS profiles of (c) L8-BO neat films and (d) D18/L8-BO blend films, respectively. (e) 2D GIWAXS profiles of D18/L8-BO active layers processed without/with different 2XBT additives.

3 | Conclusion

Three halogenated SMS additives named 2*X*BT (including 2*F*BT, 2*Cl*BT, and 2*Br*BT) were developed by attaching different halogen atoms onto their parental BT unit, to optimize the active layer morphology of the LBL-processed D18/L8-BO. Among them, 2*Br*BT offers the strongest intermolecular interactions with active layer materials of both D18 and L8-BO. Consequently, the 2*Br*BT treated D18/L8-BO film obtained superior morphology with enhanced intermolecular packing and optimized vertical phase separation, which is beneficial for improving exciton dissociation and charge transport. As a result, the 2*Br*BT processed binary OSCs provided an excellent PCE of 19.15%, outperforming these of ones treated with additive-free (18.31%), 2*F*BT (18.78%), and 2*Cl*BT (18.63%). Introducing a near-infrared absorbing eC9-4ClO guest acceptor into D18/L8-BO host to broaden absorption, the 2*Br*BT treated ternary OSCs achieved a further boosted PCE of 20.12% with a significantly improved J_{SC}. Notably, the PCE of 20.12% is the highest value among the reported LBL-processed OSCs to date. Overall, our work demonstrates that halogenation in the BT derived SMS additives is a promising strategy for optimizing active layer morphology and therefore achieving efficient OSCs.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.

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